

From Ratings to Returns: Peer Information, Farmer Beliefs, and Technology Adoption under Performance Uncertainty

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Abstract

Many smallholder farmers in low- and middle-income countries do not adopt modern agricultural inputs sold by agro-dealers due to uncertainty about their performance. We introduce a system that aggregates farmers' evaluations of seed into ratings and disseminates them to farmers and agro-dealers, providing a summary signal of seed performance based on peers' experiences. Such a system could affect outcomes through three channels: correcting farmers' beliefs and reducing their uncertainty about seed performance (belief updating), helping farmers identify better dealers (dealer sorting), or inducing dealers to improve seed preservation (supply-side accountability). We test this approach in a randomized controlled trial involving 3,500 farmers and 350 agro-dealers in the market for high-yielding maize seed in Uganda. We find that the intervention increases farmers' perceptions of seed performance, leads to higher adoption of high-yielding maize seed, and raises yields by 13%. While there is some evidence of increased switching between dealers, this switching is not directed toward higher-rated dealers, and we find no improvement in dealer-level seed performance, providing little support for dealer sorting or supply-side accountability. The results point to belief updating as the primary mechanism. Consistent with this interpretation, effects on perceptions are larger among farmers without prior purchasing experience — those with the greatest uncertainty about seed performance.

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1 Introduction

Smallholder farmers in low- and middle-income countries adopt productivity-enhancing agricultural inputs, such as improved seeds and inorganic fertilizers, at persistently low rates despite their potential to increase agricultural productivity, improve food security, and reduce poverty. As reviewed by [Suri and Udry \(2022\)](#), proposed explanations include credit, liquidity, and savings constraints; limited access to insurance; information frictions; high transaction costs and poor infrastructure; and imperfections in labor and land markets.

An important additional explanation is that farmers are uncertain about the performance of agricultural inputs supplied by agro-dealers, the primary distribution channel for improved seed varieties ([Michelson et al., 2021, 2023](#); [Dar et al., 2023](#)). For instance, in the market for high-yielding maize seed in Uganda, several studies document substantial performance uncertainty:¹ [Bold et al. \(2017\)](#) find that locally sold hybrid seed contains less than 50% authentic seed. [Barriga and Fiala \(2020\)](#) document improper handling and storage of maize seed purchased from retailers, while [Ilukor et al. \(2025\)](#) show that seed quality declines along the supply chain relative to samples obtained directly from seed companies. These studies document substantial uncertainty about the performance of seed sold through agro-dealers, but this performance is largely invisible to farmers at the point of purchase.

Performance uncertainty also distorts the use of complementary inputs: [Bulte et al. \(2025\)](#) show that farmers who are unsure about input quality allocate less labor, reducing yields and making it harder to learn about the true profitability of improved varieties. More broadly, learning from own experience is difficult because agricultural outcomes depend on many factors beyond seed quality. [Laajaj and Macours \(2026\)](#) show that agricultural learning is inherently multidimensional and that even with intensive exposure, peer-to-peer transfer of knowledge about input performance faces severe frictions. These learning constraints allow poorly informed or biased beliefs to persist.

Credible signals that reduce performance uncertainty could address this, but existing mechanisms to signal seed performance at agro-dealers have major limitations. Seed companies signal quality through labeling and packaging, but farmers struggle to identify low-quality seed without specific training ([Gharib et al., 2021](#)). Regulatory certification systems are costly and, in Uganda, do not seem to provide a reliable signal of performance ([Tyack et al., 2026](#)). Moreover, these signals only reach farmers who already engage with agro-dealers, and even certified seed may perform poorly where handling and storage are inadequate ([Barriga and Fiala, 2020](#)).

In this paper, we test a decentralized alternative: we aggregate the experiences of farmers themselves into a signal of seed performance. Building on a well-established literature documenting the importance of peers in agricultural technology adoption

¹In this study, the term “high-yielding maize varieties” refers to both open-pollinated varieties (OPVs) and hybrids that have been genetically improved through breeding programs. We explain the terminology and our rationale for using this terminology in Appendix A.1.

(Foster and Rosenzweig, 1995; Bandiera and Rasul, 2006; Conley and Udry, 2010), we test whether aggregating peers’ experiences into a public signal of seed performance can improve market outcomes through three channels: belief updating about agro-dealer seed relative to outside options, dealer sorting conditional on purchasing from agro-dealers, or supply-side improvements in seed preservation. We introduce a system that collects farmers’ evaluations of maize seed purchased from agro-dealers and aggregates them into ratings that summarize the experiences of multiple farmers facing similar conditions. These ratings are disseminated to both farmers and agro-dealers, providing a summary signal about seed performance at agro-dealers that individual farmers may find difficult to obtain through their own experimentation or social networks.

Aggregated peer ratings address key limitations of existing approaches. They are based on farmers’ evaluations of experienced seed performance and therefore reflect performance under local conditions. They are disseminated directly to farmers, allowing the information to reach those who do not typically visit agro-dealers. The system is decentralized, relying less on institutional capacity. Importantly, the system does not measure or certify objective seed quality. It aggregates farmers’ assessments of experienced performance — whether the seed they purchased yielded well, germinated reliably, and tolerated local conditions. These assessments may diverge from laboratory measures of seed purity or authenticity.

A rating system that aggregates and disseminates farmers’ purchasing experiences could improve outcomes through several channels. First, farmers may hold imprecise or biased beliefs about seed performance. Disseminating aggregated peer experiences could reduce uncertainty among all farmers and, when priors are pessimistic relative to the signal, raise expected performance among farmers without prior purchasing experience, increasing adoption.

Second, dealer-specific ratings could enable dealer sorting by revealing performance differences across sellers conditional on purchasing from agro-dealers. In agricultural input markets, farmers typically choose among a small number of local dealers with limited information about relative seed performance. When dealer-choice frictions are high relative to quality variation, consumers settle for suboptimal matches (Stigler, 1961). By aggregating dispersed information about seller-specific performance, ratings enable the kind of sorting that characterizes online review platforms (Dranove and Jin, 2010; Luca, 2016). If performance varies substantially across dealers, ratings could induce farmers to switch to higher-performing sellers.

Third, the rating system could improve dealer-level seed performance through supply-side accountability. In the prevailing market equilibrium, agro-dealers may have weak incentives to invest in seed storage and handling because farmers cannot credibly assess seed performance at the point of purchase and may struggle to attribute later outcomes such as germination or yield to dealer behavior. By aggregating farmer experiences into visible ratings, the system creates bottom-up pressure on dealers: poor seed performance becomes reputationally costly, while better-preserved seed can attract future demand. The rating system can therefore shift competition toward quality by making dealer-level performance more observable to potential buyers

(Hasanain, Khan, and Rezaee, 2023; Annan, 2025). These three channels are not mutually exclusive, but they generate distinct empirical predictions that allow us to assess their relative importance.

We test this approach in a randomized controlled trial (RCT) involving approximately 3,500 smallholder farmers and 350 agro-dealers in the market for high-yielding maize seed in eastern Uganda. The randomized design allows us to estimate causal effects, while three rounds of detailed data enable us to understand how and why the intervention affects behavior. We track outcomes along the full causal chain, from farmers’ perceptions and adoption decisions to agricultural productivity. Additional data on farmers’ choice of agro-dealers, agro-dealer seed handling practices, and objective indicators of seed performance allow us to trace the channels through which the intervention operates; measures of dealer services and signaling provide secondary evidence on other dealer responses.

Exposure to peer ratings improves farmers’ perceptions of seed performance and increases adoption of high-yielding maize seed. Treated farmers achieve 13% higher yields, showing that the intervention has economically meaningful downstream effects. Because adoption and productivity effects are consistent with all three channels, channel identification rests on the outcomes that differ across the model’s predictions. While there is some evidence that the intervention induces farmers to switch between sellers, this switching is not directed toward higher-rated dealers, providing little support for dealer sorting as a primary channel. Nor do we find improvements in dealer-level seed performance, providing little support for supply-side accountability as a primary channel. Consistent with belief updating, effects on perceptions are larger among farmers without prior purchasing experience.

This paper contributes to the literature on agricultural technology adoption, particularly on the role of input quality uncertainty (Michelson et al., 2023) and farmers’ beliefs (Michelson et al., 2021; Hoel et al., 2024). We advance this literature in three ways. First, we introduce an intervention that addresses limitations of existing seed performance signals — it is decentralized, experience-based, and reaches farmers who do not engage with formal markets — and show that it improves adoption and productivity, with the pattern of results pointing to belief updating as the primary channel. Second, by tracking outcomes on both sides of the market — farmers’ beliefs and switching behavior, and agro-dealers’ seed handling practices and seed performance — we can assess the relative importance of belief updating, dealer sorting, and supply-side accountability. Together with Dar et al. (2023), our study is among the few to connect farmer and agro-dealer perspectives, providing a more complete picture than studies focusing on buyers or sellers alone. Third, the pattern of results — positive effects on perceptions and adoption, undirected switching and no improvement in dealer-level seed performance — provides evidence that uncertainty about seed performance is a binding constraint on adoption in seed markets. While overly pessimistic or imprecise beliefs have been documented in fertilizer markets (Michelson et al., 2021; Hoel et al., 2024), we provide evidence of this mechanism in seed markets.

More broadly, this paper connects to a growing literature on crowd-sourced quality

information, transparency, and search frictions in markets with hard-to-observe seller quality. Research on online review platforms documents that consumer feedback affects both sides of the market: buyers use reviews to make more informed choices, while sellers adjust their behavior in response to public quality signals (Bajari and Hortagsu, 2004; Jin and Kato, 2006; Dranove and Jin, 2010; Reimers and Waldfogel, 2021). Related work shows that reducing search and information frictions can improve matching, trade, and market integration in both online and low-income-country markets (Bai et al., 2020; Bergquist, McIntosh, and Startz, 2024; Startz, 2024; Cai, Lin, and Szeidl, 2024; Dillon, Aker, and Blumenstock, 2025). In many of these settings, an important channel is sorting: information helps buyers identify better sellers among many options. Our study shows that in a very different setting — a low-information, offline agricultural input market — crowd-sourced ratings need not operate primarily through seller choice. In our setting, the evidence points less to sorting across agro-dealers and more to belief updating about whether agro-dealer seed performs well relative to outside options. Hasanain, Khan, and Rezaee (2023) find a related null effect on switching in Pakistan, where crowd-sourced information on livestock insemination services does not induce farmers to switch providers.

The remainder of the paper is organized as follows. Section 2 presents the conceptual model, Section 3 describes the experimental design, Section 4 outlines the empirical strategy, Section 5 describes the data, Section 6 presents the results, and Section 7 concludes.

2 Conceptual model

This section develops a simple model of how aggregated peer information affects farmers and agro-dealer behavior and outcomes. The model generates a set of predictions that helps us organize the empirical analysis.

2.1 Setup

We assume that smallholder farmers choose between purchasing higher-yielding seed varieties from agro-dealers and relying on outside options such as farmer-saved seed or local seed obtained through informal exchanges. Let θ_j denote the performance of seed sold by agro-dealer j relative to farmer i 's outside option. There are J agro-dealers in farmer i 's area, each with performance θ_j . This is the actual performance the farmer would realize if they planted the seed — it is unobservable to the farmer at the time of purchase. We decompose θ_j as $\theta_j = \theta_0 - d(e_j)$, where θ_0 is the intrinsic performance of the seed upon arrival at agro-dealer j — determined upstream in the supply chain and outside the agro-dealer's control and assumed to be the same for all agro-dealers²

²This assumption abstracts from upstream variation in the seed received by agro-dealers. Allowing θ_0 to vary by dealer would introduce an additional source of dealer heterogeneity, but would not change the model's distinction between belief updating, dealer sorting, and supply-side preservation effort.

— and $d(e_j) \geq 0$ captures performance degradation due to handling and storage, with $d'(e_j) < 0$. Higher agro-dealer effort e_j (e.g., proper storage containers, ventilation, temperature and moisture control) reduces degradation, preserving seed performance closer to its intrinsic level θ_0 .

Because seed performance is unobservable at the time of purchase, the decision to buy seed depends on beliefs about the performance of agro-dealer seed relative to these alternatives $E_i[\theta_j]$. Seed has properties of a credence good: because yields depend on many factors beyond the seed itself, farmers cannot fully assess seed performance even after use (Laajaj and Macours, 2026). Individual learning about seed is therefore noisy, slow, and difficult to self-correct (Ahmad, 2022). This is compounded by the disproportionate salience of negative information (Rozin and Royzman, 2001; Vaish, Grossmann, and Woodward, 2008; Hornik et al., 2015), which may push beliefs below the true θ . Evidence from fertilizer markets (Michelson et al., 2021; Hoel et al., 2024) and from maize seed in Uganda (Barriga and Fiala, 2020) is consistent with this hypothesis. That said, seed also has an experience good component (Foster and Rosenzweig, 1995): farmers who have purchased and planted agro-dealer seed can form more accurate beliefs than those who have not.

Farmers are risk-averse and face a two-part decision: whether to purchase seed from any agro-dealer rather than relying on the outside option (extensive margin), and if so, from which agro-dealer (agro-dealer choice). Farmer i evaluates each agro-dealer j by comparing the expected performance of that agro-dealer’s seed, discounted for uncertainty, against the cost of purchasing. Formally, the farmer purchases from the best available agro-dealer if the certainty equivalent exceeds the cost:

$$\max_j E_i[\theta_j] - \frac{\rho}{2}\sigma_{ij}^2 - c_j > 0 \quad (1)$$

where σ_{ij}^2 is farmer i ’s subjective uncertainty about agro-dealer j ’s seed performance, $\rho > 0$ is the coefficient of absolute risk aversion, and $c_j > 0$ is the cost of purchasing from agro-dealer j relative to the outside option. Adoption depends on both the level of expected performance and subjective performance uncertainty: even at the same expected return, greater uncertainty discourages adoption.

2.2 Belief updating

The intervention can affect farmer outcomes through farmers’ beliefs about θ_j . The farmer does not observe θ_j ; instead, they hold a belief $\mu_{ij} \equiv E_i[\theta_j]$ about each agro-dealer’s seed performance. Because individual learning about seed performance is difficult, beliefs are likely to be imprecise or biased, and aggregated peer information can shape belief formation.

Farmer i holds a prior belief μ_{ij} about each agro-dealer j : $\theta_j \mid \mathcal{I}_i \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2)$. In our model, the way it does so depends on farmers’ prior purchasing experience, which generates predictions about heterogeneous treatment effects. In particular, we distinguish two types based on prior purchasing experience:

- **Purchasers** ($i \in P$) have prior beliefs informed by own purchasing experience; as a result, μ_{Pj} is close to θ_j .
- **Non-purchasers** ($i \in NP$) lack own purchasing experience: we assume $\sigma_{NPj}^2 > \sigma_{Pj}^2$ (greater uncertainty), and allow μ_{NPj} to differ from θ_j . In the empirically relevant case where priors are pessimistic, $\mu_{NPj} < \theta_j$.

This captures the idea that, even though learning by doing is noisy, farmers with experience purchasing seed from agro-dealers have more precise priors.

The intervention provides farmer i with a signal about the performance of seed at agro-dealer j . The signal aggregates the experiences of n_j farmers who purchased from agro-dealer j :

$$s_j = \frac{1}{n_j} \sum_{k=1}^{n_j} (\theta_j + \varepsilon_k) = \bar{\theta}_j + \bar{\varepsilon}_j \quad (2)$$

where ε_k is the noise in rater k 's (drawn from the pool of previous purchasers) individual experience (reflecting weather, soil, management, and other factors beyond seed quality) and $\bar{\varepsilon}_j \sim \mathcal{N}(0, \sigma_\varepsilon^2/n_j)$.

To characterize how such a signal changes beliefs, we use the standard conjugate Normal-Normal Bayesian updating benchmark. Given farmer i 's prior belief about agro-dealer j 's seed performance and the signal s_j , posterior mean and variance are:

$$\tilde{\mu}_{ij} = (1 - \alpha_{ij})\mu_{ij} + \alpha_{ij}s_j, \quad \tilde{\sigma}_{ij}^2 = \frac{\sigma_{ij}^2 \cdot \sigma_\varepsilon^2/n_j}{\sigma_{ij}^2 + \sigma_\varepsilon^2/n_j} \quad (3)$$

with the weight on the signal, $\alpha_{ij} = \frac{\sigma_{ij}^2}{\sigma_{ij}^2 + \sigma_\varepsilon^2/n_j}$, increasing in prior subjective uncertainty. Since $\sigma_{NPj}^2 > \sigma_{Pj}^2$, non-purchasers place more weight on the signal and experience larger reductions in posterior uncertainty. The belief shift $\tilde{\mu}_{ij} - \mu_{ij} = \alpha_{ij}(s_j - \mu_{ij})$ is upward when the signal is more favorable than the prior. For purchasers, the main effect is a precision gain (lower $\tilde{\sigma}_{ij}^2$); for non-purchasers, the model predicts a larger precision gain and, when their priors are pessimistic relative to the signal, a larger upward revision in expected performance.

Because raters are drawn from previous purchasers, the signal need not be representative of the experience that every farmer would have with agro-dealer j . Raters may differ from non-purchasers not only in prior beliefs, but also in other dimensions such as land quality, management practices, and complementary input use. We therefore interpret s_j as an informative peer-experience signal rather than an unbiased estimate of a universal agro-dealer-specific treatment effect. The model requires that, conditional on the local market and agro-dealer, average rater experiences contain information about agro-dealer seed performance beyond selection-driven differences across farmers. Under this condition, averaging across raters reduces idiosyncratic noise: more raters produce a more precise signal about agro-dealer j 's seed performance. Separately, the Bayesian updating expression should be read as a benchmark in which farmers perceive the rating signal as credible and relevant. Lower credibility or attention would attenuate the

weight placed on s_j , without changing the qualitative prediction that more uncertain priors update more strongly.

2.3 Dealer sorting

While the belief-updating channel captures changes in farmers' beliefs about agro-dealer seed relative to outside options, dealer sorting captures changes in the choice of seller conditional on purchasing from agro-dealers. Because the intervention provides agro-dealer-specific ratings (s_j), it conveys two types of information. First, the average signal across agro-dealers, $\bar{s} = \frac{1}{J} \sum_j s_j$, provides information about the general performance of agro-dealer seed, allowing belief updating about agro-dealers overall. Second, the variation in signals across agro-dealers, $s_j - \bar{s}$, reveals performance differences and enables sorting.

Whether sorting occurs depends on the dispersion in θ_j across agro-dealers. If performance varies substantially across agro-dealers and ratings reveal this variation, farmers can switch to higher-rated agro-dealers — predicting increased switching. If, however, ratings are similar across agro-dealers, the signal contains little agro-dealer-specific information, and the sorting channel is weak.

2.4 Supply-side accountability

The intervention can also affect farmer outcomes by inducing agro-dealers to reduce degradation $d(e_j)$, raising θ_j itself. This is a supply-side accountability channel: by aggregating farmers' experiences into a visible market signal, the rating system raises the reputational cost of selling poorly preserved seed or exerting low effort in seed storage and handling. When ratings are disseminated to farmers, agro-dealers' performance becomes visible to potential buyers: their customers' experiences are collected, summarized, and shared.

Let $e_j \geq 0$ denote agro-dealer j 's effort in seed preservation (e.g., proper storage containers, ventilation, temperature and moisture control), with convex cost $c(e_j)$. Recall that $\theta_j = \theta_0 - d(e_j)$ with $d'(e_j) < 0$: higher effort reduces degradation, preserving seed performance closer to θ_0 . Without the intervention, agro-dealers' reputational incentives are limited to repeat purchases and informal word of mouth. Let $R_C(\theta_j)$ denote the baseline demand for agro-dealer j that depends on quality through these informal channels, with $R_C' > 0$. Under the intervention, ratings are collected and disseminated, amplifying the reputational return to quality. Let $R_T(\theta_j)$ denote demand under the rating system, with $R_T'(\theta_j) > R_C'(\theta_j)$ — the marginal return to quality is higher when performance is publicly visible. The agro-dealer chooses effort to maximize:

$$\max_{e_j} R_\tau(\theta_0 - d(e_j)) - c(e_j), \quad \tau \in \{C, T\} \quad (4)$$

The first-order condition is $R_\tau'(\theta_0 - d(e_j^*))(-d'(e_j^*)) = c'(e_j^*)$. Since $R_T' > R_C'$, optimal effort is higher under the intervention: $e_{jT}^* > e_{jC}^*$. If agro-dealers respond to the

intervention by investing in seed preservation, degradation falls and seed performance improves. If both channels operate, they may reinforce each other: belief updating increases farmers’ willingness to engage with agro-dealers, while better seed preservation strengthens the actual purchasing experience.

2.5 Testable predictions

The model identifies three channels through which the intervention could affect outcomes: belief updating (Section 2.2), dealer sorting (Section 2.3), and supply-side accountability (Section 2.4). Each channel generates distinct predictions across five outcomes, summarized in Table 1.

- **Prediction 1 (Perceptions).** Under the belief updating channel, the treatment changes perceived seed performance in the direction of the signal, with stronger effects among non-purchasers. If prior beliefs are pessimistic relative to the signal, perceived seed performance shifts upward. Under the dealer sorting channel, there is no direct prediction on perceptions.³ Under the supply-side accountability channel, perceptions may improve after dealer behavior changes and improved performance becomes observable, but such effects need not be concentrated among non-purchasers.
- **Prediction 2 (Adoption).** All three channels can increase adoption under the maintained assumption that ratings reduce decision-relevant uncertainty, reveal higher-performing dealers, or induce improvements in seed performance. Belief updating can increase adoption by reducing uncertainty and, when priors are pessimistic relative to the signal, by raising expected performance above the adoption threshold. Dealer sorting can increase adoption if identifying better dealers makes agro-dealer seed more attractive relative to outside options, while supply-side accountability can increase adoption by improving the objective quality of seed on offer. Adoption therefore does not discriminate sharply between channels.
- **Prediction 3 (Productivity).** All three channels can raise yields if they increase adoption of genuinely higher-performing seed or improve the performance of seed actually purchased. Dealer sorting and supply-side accountability may additionally raise yields conditional on adopting — through sorting to better agro-dealers and through improved seed quality, respectively. Productivity therefore does not discriminate sharply between channels. For belief updating, yield gains require that, on average, peer ratings move adoption decisions toward seed with higher true performance relative to outside options.
- **Prediction 4 (Switching).** The three channels differ in their predictions about switching between agro-dealers. Under the belief-updating channel as defined

³If farmers sort to better dealers and subsequently experience higher seed performance, perceptions could improve indirectly over time. However, this would be a downstream consequence of switching behavior rather than an immediate effect of receiving ratings.

here, peer ratings primarily shift beliefs about agro-dealer seed relative to outside options rather than reallocating farmers across specific agro-dealers; by itself, this channel therefore does not predict directed switching toward higher-rated dealers. Under the dealer sorting channel, dealer-specific ratings reveal performance differences, enabling farmers to switch to higher-rated dealers, predicting increased switching. The supply-side accountability channel has no direct prediction on switching. Finding that any increase in switching is not directed toward higher-rated dealers weakens evidence for dealer sorting as the primary channel.

- **Prediction 5 (Seed performance).** The three channels differ in their predictions about objective seed performance. Under belief updating, the intervention shifts beliefs but not seed performance at a given agro-dealer: θ_j is unchanged. Under dealer sorting, the intervention may change the performance of seed actually purchased by farmers if they reallocate toward higher-performing agro-dealers, but this operates through dealer choice rather than through product improvement at any given agro-dealer. Under the supply-side accountability channel, if agro-dealers respond to reputational incentives by investing in handling and storage, degradation $d(e_j)$ falls and dealer-level seed performance θ_j improves. Finding no treatment effect on dealer-level seed performance weakens evidence for the supply-side accountability channel, subject to measurement precision and timing.

The channels may also differ in their predicted timing of effects. Belief updating and dealer sorting can affect behavior as soon as farmers receive ratings: beliefs may shift immediately, and dealer choices can adjust at the next purchase occasion. These effects may persist or evolve through endline as farmers accumulate own and peer experience. Supply-side accountability may take longer to affect farmer perceptions and adoption, because any dealer improvements in seed storage and handling must first translate into realized seed performance and then become observable to farmers through experience or updated ratings. This timing distinction is useful when interpreting whether effects appear first at midline, persist to endline, or emerge only after an additional rating cycle.

3 Experimental design

3.1 Randomization and power

Randomization is conducted at the agro-dealer catchment-area level with equal assignment probability. Catchment areas consist of geographically proximate clusters of towns, villages, markets, trading centers, and other key hubs, and typically include

Table 1: Model predictions by channel and outcome

	Belief updating	Dealer sorting	Supply-side
Perceptions	Move toward signal, larger for non-purchasers	No prediction	May improve, uniform across farmers
Adoption	Increases through lower uncertainty or higher expected performance	Increases, uniform	Increases, uniform
Productivity	Higher yields (via adoption)	Higher yields (via better matching)	Higher yields (via better seed)
Switching	No change	Increases	No prediction
Dealer-level seed performance	No change	No change	Improves ($d(e_j)$ falls)

Note: Adoption effects alone do not discriminate between channels. Heterogeneity in perceptions (P1), undirected switching (P4), and lack of dealer-level seed performance improvements (P5) jointly point to belief updating as the dominant channel.

several agro-dealers.⁴ Because the intervention changes the local information environment, treatment is assigned at the catchment-area level rather than within catchments. Farmer-level treatment effects are identified by comparing farmers in treated and control catchment areas. As described below, farmers were sampled from villages linked to agro-dealers in each catchment area.

Sample sizes were determined using simulation-based power calculations based on data from prior surveys, allowing power to be assessed under empirically relevant outcome distributions at both the farmer and agro-dealer levels. Full details are provided in the [pre-analysis plan](#) (AEA RCT Registry, RCT ID 0006361).

3.2 Intervention

Rating collection and computation

Treated farmers were asked to rate maize seed sold by agro-dealers in their catchment area. Farmers could rate an agro-dealer only if they had personally purchased seed from that agro-dealer or knew someone who had. In practice, ratings were based primarily on direct purchase experience, with indirect reports accounting for a smaller share.

⁴To define these areas, we used a haversine function to construct an adjacency matrix based on GPS coordinates, assigning agro-dealers less than five kilometers apart to the same catchment area. The five-kilometer threshold was selected based on a visual inspection of the map, the average village size in our sample, and the reported distance between farmers and agro-dealers in survey data from a previous study of the maize value chain, available [here](#).

Table 2: Questions for farmers to rate agro-dealers

	min	max
Do you know this <i>shop name</i> or <i>dealer name</i> , sometimes called <i>nickname</i> , located in <i>market name</i> ?	no	yes
The place can be described as <i>description</i> .		
Please rate this agro-dealer on:		
Quality and authenticity of seed	1 star	5 stars
Please rate the maize seed that this agro-dealer sells on:		
General quality	1 star	5 stars
Yield as advertised	1 star	5 stars
Drought tolerance as advertised	1 star	5 stars
Pest/disease tolerance as advertised	1 star	5 stars
Speed of maturing as advertised	1 star	5 stars
Germination	1 star	5 stars

Enumerators guided farmers through the list of agro-dealers using a tablet-based application displaying names, location descriptions, and storefront pictures (see Subsection 5.1).

Ratings were collected post-harvest, when farmers could form noisy assessments of seed performance based on realized outcomes such as germination, yield, drought resilience, pest and disease resistance, and crop maturation speed, and were elicited using the questions listed in Table 2 on a five-point star scale. The rating dimensions capture attributes that farmers attend to when purchasing and using seed, as identified through prior qualitative work, including workshops with experts from MAAIF, NARO, USTA, UNADA, Naseco Seed Company, and extension specialists.

Because agro-dealers stock heterogeneous seed varieties with different agronomic profiles, assessing overall seed performance in absolute terms may obscure trade-offs across characteristics. To address this, the rating instrument was designed to capture performance relative to product claims. Farmers were asked to rate attributes “as advertised,” e.g. by evaluating “maize seed sold by the agro-dealer” on “yield as advertised.” This approach allows performance to be assessed on a comparable basis across varieties while preserving an agro-dealer-level performance signal: an agro-dealer selling a variety without stress-tolerance traits that performs as advertised can receive a high rating, whereas an agro-dealer selling a stress-tolerant variety that underperforms relative to its claims receives a lower rating.

A potential concern is that the rating task itself could increase farmers’ awareness of agro-dealers, potentially confounding treatment effects. To address this, control-group farmers were also guided through the list of agro-dealers in their catchment area in a comparable manner (displaying names, location descriptions, and storefront pictures),

ensuring comparable exposure. However, they were not asked to provide ratings, as the act of rating is a component of the treatment. This design also reduces the scope for reactivity and experimenter bias.

Average ratings for each agro-dealer were computed by aggregating all farmer responses within the catchment area. Details on the rating computation are provided in Appendix A.2.

Rating dissemination to farmers

The treatment relied on timely dissemination of agro-dealer ratings before the next agricultural season, allowing farmers to incorporate this information into seed purchasing decisions. Dissemination occurred through short message service (SMS) and in-person visits.

Farmers received one SMS per agro-dealer in their catchment area, delivered in one of three local languages – Lusoga, Lugwere, or Samia. For farmers in treatment areas, the message read:

Hello from AgroAdvisor! Did you know that customers rate the quality of maize seed sold at [*name of the agro-dealer*] as [*okay/good/very good/excellent*]?

Control farmers received a placebo message that merely highlighted the existence of agro-dealers in their catchment area.

In addition to SMS dissemination, enumerators revisited farmers and displayed agro-dealer ratings using a tablet-based application. The application cycled through all agro-dealers in the catchment area (again displaying names, location descriptions, and storefront pictures), presenting this message accompanied by a one-to-five-star visual rating:

We wanted to let you know that customers rate the quality of maize seed sold at [*name of the agro-dealer*] as [*okay/good/very good/excellent*]! The maize seed from this agro-dealer received a score of [*score*] out of 5!

For control group farmers, the application similarly cycled through agro-dealers in the catchment area without displaying ratings. Together with the placebo SMS, this design isolates the effects of rating information from potential reactivity or salience effects arising from reminding farmers of the existence of agro-dealers in their area.

Rating dissemination to agro-dealers

Agro-dealers received their ratings in the form of a report. The front page displayed a certificate with a logo and the agro-dealer’s overall rating. The back page provided more detailed information, presenting ratings for seed sold by the agro-dealer across the six key dimensions and a table displaying average ratings of other agro-dealers in the same catchment area. An example of the full report can be found in Appendix A.3.

The study spans two agricultural seasons with three data collection rounds. At baseline (before season 1), we survey farmers and agro-dealers and collect ratings from treated farmers. Ratings are aggregated and disseminated to farmers and agro-dealers before planting in season 1. At midline (between seasons 1 and 2), we collect outcomes and a second round of ratings from treated farmers. Updated ratings are disseminated before planting in season 2. At endline (after season 2), we collect final outcomes.

4 Empirical strategy

We estimate intention-to-treat effects using Ordinary Least Squares:

$$Y_{ij} = \alpha + \beta T_j + \gamma' X_{ij} + \delta Y_{0ij} + \varepsilon_{ij} \quad (5)$$

where Y_{ij} is the outcome for farmer i in catchment area j , Y_{0ij} is the corresponding mean-centered baseline outcome, T_j is a binary indicator for treatment assignment, and X_{ij} includes controls for orthogonal treatments in the factorial design, as the study was embedded in a larger project with additional cross-randomized interventions (demeaned and interacted with the main treatment effect; see [Lin, 2013](#); [Muralidharan, Romero, and Wüthrich, 2023](#)). The coefficient β captures the intention-to-treat effect. We estimate analogous specifications for agro-dealer-level outcomes, where Y_{ij} denotes the outcome for agro-dealer i in catchment area j .

Since randomization occurs at the catchment area level, we cluster standard errors accordingly. For farmer-level outcomes, we use the conventional sandwich estimator without small-sample corrections. For agro-dealer-level outcomes, we apply the Bell-McCaffrey adjustment to account for the smaller number of observations ([Imbens and Kolesár, 2016](#)).

Following our pre-analysis plan, we trim continuous outcomes to limit the influence of outliers and apply inverse hyperbolic sine transformations to skewed variables. Outcomes with limited variation, defined as having 95% or more of observations at a single value within the relevant sample, are excluded from the analysis.

To account for multiple hypothesis testing, we aggregate related outcomes into summary indices following [Anderson \(2008\)](#), which assigns lower weights to highly correlated outcomes and thereby concentrates information. As a robustness check, we also compute summary indices following [Kling, Liebman, and Katz \(2007\)](#). While these indices capture overall effects across outcome families, we also report treatment effects on individual outcomes to help interpretation, with individual significance treated cautiously.

Our primary approach to interpreting mechanisms relies on the pattern of effects across outcomes: each channel generates a distinct empirical signature across perceptions, adoption, productivity, switching, and seed performance (Table 1). We additionally examine heterogeneity by prior purchasing experience. This analysis was not pre-specified and is therefore exploratory, but it is motivated by the model’s prediction that farmers with greater uncertainty about seed performance should place more weight on the peer signal.

5 Data

5.1 Sample

The agro-dealer sample was constructed by listing all agro-dealers across eleven districts in southeastern Uganda, drawing on official records of registered agro-dealers and extensive field visits to identify new and informal ones. Following this census, the 348 eligible agro-dealers were assigned to 130 catchment areas (see Subsection 3.1 for details). Of the 130 catchment areas, 65 were assigned to treatment, covering 193 agro-dealers, and 65 to control, covering 155 agro-dealers. Across all catchment areas, the average number of agro-dealers per area was three, with the number ranging from one to 18. To link agro-dealers with farmers, agro-dealers identified the villages where most of their customers reside. Enumerators then randomly sampled ten maize-growing households from these villages, yielding a sample of approximately 3,500 smallholder farmers.

Baseline data were collected from agro-dealers in September-October 2020 and from farmers in April 2021; midline data were collected in January-February 2022, and end-line data in July-August 2022. At the agro-dealer level, enumerators interviewed the individual most knowledgeable about day-to-day operations, purchased a maize seed bag, and recorded indicators related to seed storage conditions. At the farmer level, enumerators interviewed the individual most knowledgeable about maize farming. In addition to general questions, farmers listed all maize plots, from which one was randomly selected for detailed data collection, which avoids selective reporting across multiple plots and allows us to connect sourcing to plot-level inputs, effort and harvest outcomes.

5.2 Baseline descriptives and randomization balance

Table 3 reports farmer characteristics at baseline. The average household head is 49 years old, 78% are male, and 51% have completed primary education. Households have nine members on average and are located four kilometers from the nearest agro-dealer and nine kilometers from the nearest tarmac road. Farmers cultivate three acres, have 23 years of maize-growing experience, and reported yields of 440 kilograms per acre in the season preceding the baseline survey.

Half of farmers planted high-yielding maize seed on at least one plot in that season.⁵ One in three farmers purchase seed from an agro-dealer. Other farmers obtained high-yielding seed from Operation Wealth Creation/National Agricultural Advisory Services or non-governmental organizations; many relied on their own retained seed or farmer-to-farmer exchanges. These patterns support the assumption that farmers have mean-

⁵“High-yielding maize varieties” refers to both open-pollinated varieties and hybrids that have been improved through breeding programs. “Farmer-saved seed” refers to seed of unknown progeny or quality, which typically performs worse than freshly purchased high-yielding maize varieties from agro-dealers. Our terminology and the rationale for these definitions are detailed in Appendix A.1.

ingful outside options to purchasing seed from agro-dealers. At the same time, 68% of farmers report believing that maize seed sold by agro-dealers is adulterated. This widespread concern underscores that farmers perceive agro-dealer seed as risky, so credible information about seed performance could affect adoption decisions by changing either expected performance or uncertainty.

Table 4 reports baseline characteristics of agro-dealers and their shops. The average respondent is 32 years old, 60% are male, and over 90% have completed primary education. In 55% of cases, the respondent is also the owner of the agro-shop. On average, agro-shops were established five years before the baseline survey, are located seven kilometers from the nearest tarmac road, and serve 41 customers per day. 74% sell only farm inputs and equipment, 60% provide credit, and 46% offer advisory services.

With the shop manager’s permission, enumerators inspected seed storage areas and recorded observable characteristics. They reported pest issues (e.g., rats or insects) at 65% of agro-shops and maize seed stored in open containers at 16% of agro-shops. Enumerators also sampled maize seed bags at the agro-shops and measured their moisture content. Moisture is a commonly used objective indicator of seed performance because it directly affects germination and viability; additional details are provided in Appendix A.1. The average moisture content was 13.6%. Regarding packaging and labeling, 68% of purchased seed bags displayed a packaging date, 18% an expiry date, and only 8% a quality certification label issued by the National Seed Certification Services (NSCS), consistent with weak baseline guarantees of seed performance provided through agro-shops.⁶

Tables 3 and 4 also double as balance tables. In particular, we present standard orthogonality tests with pre-registered variables to assess baseline comparability between treatment and control groups. For farmer-level outcomes (Table 3), no comparison is significant; for agro-dealer-level outcomes (Table 4), two of 17 comparisons are significant at the 10% level. Joint F-tests further confirm baseline balance.

5.3 Attrition

Table 5 reports attrition rates for treatment and comparison groups. At midline, 12% of agro-dealers and 2% of farmers were missing; at endline, 14% of agro-dealers and 1% of farmers were not surveyed. To assess whether attrition was linked to treatment, we regressed the likelihood of leaving the sample on the treatment indicator and found that treated agro-dealers were significantly less likely to drop out.⁷

Differential attrition was further assessed by regressing pre-registered balance variables (see Table 4) on indicators for leaving the sample, for belonging to the control group, and on their interaction (Appendix A.4). Results indicate that attrition among

⁶In Uganda, maize seed certification is overseen by the NSCS within MAAIF. Certified seed is identified by a blue tag or sticker, while quality-declared seed carries a green tag or sticker.

⁷Reasons for agro-dealer attrition at endline were: eleven shops closed, ten relocated, eight changed product offerings, three merged with other shops, one refused to be interviewed, and 17 reported other (4) or no specific reasons (13).

Table 3: Descriptive statistics and orthogonality tests - Farmer

	mean	treated	obs.
Household head's age in years [†]	48.62 (13.38)	-0.24 (0.56)	3453
Household head is male [†]	0.78 (0.42)	0.03 (0.03)	3470
Household head finished primary education [†]	0.51 (0.50)	0.04 (0.03)	3437
Homestead's distance to nearest tarmac road in km	9.39 (10.81)	-1.23 (1.71)	3302
Homestead's distance to nearest agro-dealer in km [†]	3.78 (4.79)	0.11 (0.37)	3339
Number of people in household (incl. respondent) [†]	8.70 (3.98)	-0.09 (0.18)	3470
Number of rooms in house	3.49 (1.45)	0.02 (0.09)	3469
Farmer's land for crop production in acres	3.35 (4.32)	0.00 (0.22)	3442
Years since farmer started growing maize	23.09 (13.14)	-0.55 (0.58)	3470
Yield in kg/acre [†]	443.01 (304.99)	-6.14 (13.52)	3308
Farmer used high-yielding maize varieties on any plot [†]	0.49 (0.50)	0.01 (0.02)	3466
Farmer bought this seed at agro-dealer [†]	0.32 (0.47)	0.01 (0.02)	3424
Amount of this seed farmer bought at agro-dealer in kg [†]	9.52 (6.92)	-0.34 (0.53)	1064
Farmer thinks maize seed at agro-dealer is adulterated [†]	0.68 (0.46)	0.00 (0.03)	2673
Farmer used DAP/NPK	0.25 (0.43)	0.02 (0.04)	3465
Farmer used organic manure	0.07 (0.26)	0.01 (0.01)	3466

Note: Column (1) reports sample means at baseline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; column (3) reports the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels. [†] indicates that the variable was pre-registered to test balance and is included in the F-test.

Table 4: Descriptive statistics and orthogonality tests - Agro-dealer

	mean	treated	obs.
Respondent's age in years [†]	32.43 (11.49)	-2.24 ⁺ (1.21)	347
Respondent is male [†]	0.59 (0.49)	-0.01 (0.06)	348
Respondent finished primary education [†]	0.92 (0.27)	-0.01 (0.03)	339
Respondent owns shop	0.55 (0.50)	0.02 (0.06)	348
Respondent received training on maize seed handling [†]	0.53 (0.50)	0.12 ⁺ (0.07)	348
Respondent knows how to store seed after repackaging [†]	0.27 (0.44)	0.08 (0.06)	348
Agro-dealer's distance to nearest tarmac road in km [†]	6.56 (10.39)	-1.58 (2.24)	343
Agro-dealer only sells farm inputs	0.74 (0.44)	0.03 (0.06)	348
Years since agro-dealer establishment [†]	5.34 (6.30)	0.21 (0.78)	348
Number of customers per day [†]	41.49 (46.49)	6.43 (6.72)	346
Quantity of maize seed sold in kg [†]	695.50 (1497.18)	176.31 (235.92)	340
Amount of maize seed lost/wasted last season in kg [†]	3.50 (18.65)	2.40 (2.30)	348
Number of maize varieties in stock	2.83 (1.59)	0.19 (0.21)	344
Agro-dealer has problem with pests	0.65 (0.48)	-0.03 (0.06)	347
Agro-dealer stores maize seed in open containers	0.16 (0.36)	0.08 (0.05)	347
Agro-dealer received seed related complaint from customer	0.64 (0.48)	0.07 (0.05)	347
Moisture in bag of maize seed in %	13.56 (1.44)	-0.18 (0.26)	347

Note: Column (1) reports sample means at baseline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; column (3) reports the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels. [†] indicates that the variable was pre-registered to test balance and is included in the F-test.

Table 5: Attrition

	<i>midline</i>		<i>endline</i>	
	mean	treated	mean	treated
Agro-dealer left the sample	0.121 (0.326)	-0.108** (0.035)	0.144 (0.351)	-0.079+ (0.042)
Farmer left the sample	0.018 (0.134)	0.001 (0.005)	0.008 (0.091)	-0.001 (0.003)

Note: Columns (1) and (3) report sample means at mid- and endline and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; **, *, and + denote significance at the 1%, 5%, and 10% levels.

agro-dealers is unlikely to bias our estimates. No differential attrition was observed at the farmer level; since primary outcomes are measured among farmers, attrition is unlikely to meaningfully affect the main treatment effects.

5.4 Rating data

At endline, the rating model generated a dyadic rating dataset comprising of 16,823 farmer-dealer links. Among farmers who provided at least one rating, the average farmer rated 1.4 agro-dealers (min = 1, max = 10). The average agro-dealer was rated by 6.7 farmers (min = 0, max = 30).

Because ratings could be provided either by farmers who had personally purchased seed from the agro-dealer or by those who knew someone who had, the signal may reflect indirect information rather than firsthand performance experience. In practice, 93% of ratings were provided by farmers with direct purchase experience, and ratings are nearly identical between purchasers and non-purchasers (see Appendix A.5 for details). This suggests that the aggregate signal predominantly reflects direct experience.

Ratings may also reflect factors unrelated to seed performance, such as customer service or personal relationships. Appendix A.5 provides suggestive evidence that ratings are associated with observable characteristics plausibly related to seller and seed performance: specialized agro-shops and dealers with better seed handling practices receive higher ratings, and seed ratings on randomly selected fields are positively correlated with maize yields. These patterns are consistent with ratings reflecting farmers' assessments of seed performance rather than purely non-performance attributes. Appendix A.5 provides additional detail on the rating data, including the distribution of ratings across the six performance dimensions and their inter-correlations.

6 Results

We report treatment effects on outcomes for both farmers and agro-dealers at midline (after one agricultural season) and endline (after two seasons). All outcome variables were pre-specified in a registered pre-analysis plan at the [AEA’s registry for RCTs](#). The full econometric analysis was run on simulated data prior to outcome data collection and documented in a [registered report](#) ([Humphreys, De la Sierra, and Van der Windt, 2013](#)). All documents, code, and data are publicly available in a version-controlled [GitHub repository](#).⁸

6.1 Perceived seed performance

We begin with the first link in the causal chain: whether the clearinghouse shifts farmers’ perceptions of agro-dealer seed quality. Table 6 reports treatment effects on a perception index that combines farmer ratings of agro-dealer seed performance across six dimensions: general quality, yield, drought tolerance, pest and disease tolerance, maturity time, and germination (see Table 2 for the exact questions used for measuring perceptions). The clearinghouse increases this index by 0.09 standard deviations, significant at the 10% level. The positive sign is consistent with the belief-updating channel in the model if the peer signal was more favorable than farmers’ prior beliefs about agro-dealer seed performance.

These estimates should be interpreted with some caution. The perception index requires non-missing ratings on all six dimensions for at least one dealer, and ratings were collected from farmers in control catchment areas only at endline. Treatment effects on perceptions can therefore be estimated only for the final survey round and are less precise than estimates for the main adoption and productivity outcomes.

The model also predicts stronger updating among farmers with greater prior uncertainty. We examine this prediction using prior agro-dealer seed-purchasing experience, proxied by whether the farmer used improved maize seed from an agro-dealer in the season preceding baseline. The subgroup estimates are similar: 0.102 standard deviations among farmers without prior experience and 0.092 among prior purchasers, and a formal interaction test does not reject equality. We therefore interpret the perception results as evidence of upward belief updating overall, rather than as strong evidence of differential updating by prior experience.

6.2 Adoption and sourcing decisions

Table 7 reports treatment-control estimates of the clearinghouse effects on farmers’ adoption and sourcing decisions. Consistent with the model’s prediction that credi-

⁸The presentation of results differs slightly from the pre-registered structure due to a reordering of outcomes, which modifies the construction of some pre-specified indices but leaves the main conclusions unchanged. Analyses adhering strictly to the pre-analysis plan remain publicly available in the project repository (see, e.g., the archived version linked [here](#)).

Table 6: Treatment effects on farmers’ perceived seed performance

	<i>baseline</i>	<i>full sample</i>		<i>sub-sample</i>	
	mean	treated	obs.	treated	obs.
Index of farmer’s seed ratings of agro-dealers ¹	0.000 (0.637)	0.092+ (0.054)	1664	0.102+ (0.055)	996

Note: Column (1) reports the baseline mean and standard deviation below; column (2) reports the difference between treatment and control group and the standard error below; it is clustered at the level of randomization; column (3) reports the number of observations; columns (4) and (5) mirror this structure for the sub-sample of farmers who did not use high-yielding maize seed purchased from an agro-dealer on any plot at baseline, **, *, and + denote significance at the 1%, 5%, and 10% levels; larger indices indicate more desirable outcomes.

¹The index of farmer’s maize seed ratings contains 6 ratings: general quality, yield, drought tolerance, pest/disease tolerance, time of maturity, germination. The ratings are aggregated at farmer level (one farmer rates multiple agro-dealers), then the index is computed.

ble performance information can increase adoption by reducing decision-relevant uncertainty, the [Anderson \(2008\)](#) index of adoption outcomes is significantly higher for treated farmers than for control farmers at both midline and endline ($p < 0.05$). The response is concentrated on the extensive margin: treated farmers are more likely to source maize seed from agro-dealers, while quantities purchased conditional on buying do not differ significantly between treated and control farmers.

Across all plots, treated farmers are six percentage points more likely than control farmers to purchase high-yielding maize seed from an agro-dealer at midline, equivalent to 18% of the baseline mean. By endline, the estimated agro-dealer sourcing effect is three percentage points and is no longer statistically significant, although treated farmers remain more likely than control farmers to adopt improved maize varieties in both survey rounds: 3.5 percentage points at midline and 4.2 percentage points at endline.

We next examine the same sourcing margin on a randomly selected maize plot. Treated farmers are nearly five percentage points more likely at midline and 3.6 percentage points more likely at endline to obtain seed for the randomly selected plot from an agro-dealer. Seed expenditure on the plot is also significantly higher for treated farmers than for control farmers. These treatment-control differences are accompanied by reduced reliance on farmer-saved seed at midline. Estimated effects on individual hybrid and OPV variety categories are positive but imprecise.⁹ These results indicate that the intervention shifted farmers toward agro-dealer seed, but they do not by themselves distinguish whether this reflects belief updating, dealer sorting, or supply-side responses.

⁹We classify varieties as improved if they correspond to recognized hybrid or OPV types and the seed was newly purchased rather than recycled or farmer-saved.

Table 7: Treatment effects on farmers' adoption and sourcing decisions

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Farmer bought seed at agro-input shop for any plot [†]	0.325 (0.468)		0.059** (0.021)	3145	0.031 (0.020)	3225
Amount of this seed farmer bought at agro-input shop in kg	9.519 (6.920)		-0.105 (0.358)	599	0.378 (0.431)	621
Farmer used high-yielding seed on any plot [†]	0.492 (0.500)		0.035+ (0.020)	3206	0.042* (0.020)	3282
Farmer bought seed at agro-input shop for specific plot [†]	0.330 (0.470)		0.047* (0.022)	3153	0.036+ (0.019)	3240
Farmer used farmer-saved seed on specific plot	0.579 (0.494)		-0.042+ (0.022)	3153	-0.016 (0.020)	3240
Cost of seed used on specific plot in UGX ^{§†}	14078.272 (24654.685)		0.499* (0.235)	2848	0.350+ (0.209)	2942
Farmer used hybrid seed/OPV on specific plot ^{1†}	0.432 (0.495)		0.035 (0.023)	2954	0.030 (0.023)	3047
Overall Anderson index	-0.013 (0.899)		0.087* (0.042)	2854	0.086* (0.039)	2978
Max. number of obs.				3407		3441

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; [†] indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

¹For this variable, only non-recycled (newly purchased, not farmer-saved) seed counted as hybrid variety/OPV.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

We next examine whether increased farmer engagement is reflected at the agro-dealer level and consider various agro-business related outcomes (Table 8). The outcomes include self-reported sales volume for the four most common high-yielding maize varieties in the previous season (Longe 7H, Longe 10H, Longe 4, and Longe 5), average start-of-season prices, implied revenue, the number of maize-seed customers served per day, and the number of maize varieties stocked.

The agro-dealer level results are consistent with increased market activity observed at the farmer level. At midline, treated agro-dealers score significantly higher on the [Anderson \(2008\)](#) index, and their estimated revenue is nearly 20% higher than that of control agro-dealers, significant at the 10% level. This revenue effect appears more consistent with expanded sales than price increases, though the volume and price components are not individually statistically significant. At endline, the treatment effect on the business-activity index is significant at the 1% level, driven primarily by treated agro-dealers reporting 31% more customers than control agro-dealers, or roughly six additional customers per day relative to the baseline mean.

Taken together, the farmer and dealer results suggest that the clearinghouse increased engagement with agro-dealer seed markets. Treated farmers became more likely than control farmers to purchase high-yielding seed from agro-dealers, and treated agro-dealers report higher customer activity and revenue than control agro-dealers. These effects are consistent with the model’s adoption prediction, but because adoption and market activity can arise under all three channels, they are interpreted as downstream market responses rather than as evidence for a specific mechanism. Appendix A.6 reports business outcomes for two recently released maize varieties (Longe 10H and Longe 5).

6.3 Harvest outcomes

We examine harvest-related outcomes on a randomly selected maize plot in Table 9. Outcomes include production, plot size, yield, the value of maize production at market prices net of seed costs, maize retained for seed in the following season, and maize sales outcomes.

Results show positive endline effects. The overall index is positive and statistically significant, and treated farmers achieve yields that are 56 kilograms per acre higher than the control group, a 13% increase relative to the baseline mean and significant at the 1% level. The value of maize production net of seed costs is 66,000 UGX higher for treated farmers, approximately 18 USD at the time of the survey. Treated farmers also retain less maize for seed at midline, consistent with reduced reliance on farmer-saved seed. We find no effects on quantity sold or total maize revenue. This suggests that the yield gains did not translate into detectable increases in marketed surplus, and that additional output was likely retained within the household.

The harvest effects provide downstream evidence that the intervention improved realized agricultural outcomes, not only beliefs or sourcing decisions. They are consistent with the model’s prediction that credible seed-performance information can raise yields

Table 8: Treatment effects on agro-dealers' business activity

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Maize seed sales volume in kg ^{§†}	695.503 (1497.183)		0.284 (0.227)	292	0.239 (0.253)	286
Sales price of maize seed in UGX/kg [†]	4273.897 (955.073)		99.272 (113.292)	275	145.861 (138.816)	264
Revenue from maize seed in mln UGX ^{§†}	2.890 (6.286)		0.185 ⁺ (0.108)	292	0.143 (0.118)	286
Number of maize seed customers per day ^{§†}	19.764 (20.689)		0.127 (0.101)	294	0.310 ^{**} (0.112)	288
Number of maize varieties in stock [†]	2.834 (1.589)		0.245 (0.245)	295	0.221 (0.220)	292
Overall Anderson index	0.031 (0.610)		0.197 [*] (0.092)	274	0.238 ^{**} (0.082)	270
Max. number of obs.				306		297

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; [†] indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

when it shifts farmers toward higher-performing seed or improves the performance of seed actually purchased. However, productivity effects do not distinguish sharply between belief updating, dealer sorting, and supply-side accountability, since all three channels can generate yield gains.

A back-of-envelope calculation suggests that the adoption margin alone is unlikely to mechanically account for the full yield effect. Even assuming improved varieties yield 50 percent more than farmer-saved seed, the roughly 4 percentage point increase in agro-dealer sourcing would explain only a modest share of the observed 56 kg/acre yield gain. The remaining effect could reflect changes in how farmers use purchased seed. By reducing uncertainty about seed performance, the clearinghouse may increase incentives to invest in complementary inputs whose returns depend on seed quality, as in [Bulte et al. \(2025\)](#). Consistent with this explanation, treated farmers are approximately 5 percentage points more likely to use inorganic fertilizer at both midline and endline and are less likely to intercrop at midline.

6.4 Alternative channels: dealer sorting and supply-side accountability

The results so far are consistent with belief updating: the clearinghouse improves perceived seed performance and increases engagement with agro-dealer seed markets. However, as outlined in our model in Section 2, the downstream adoption and productivity effects could also arise through two other channels in the model. First, ratings may help farmers sort across agro-dealers. Second, public ratings may increase agro-dealers' incentives to improve the performance of the seed they sell. We therefore turn to these two alternative channels.

6.4.1 Dealer sorting and agro-dealer switching

Because the clearinghouse provides agro-dealer-specific ratings, it may help farmers identify higher-performing sellers and reallocate purchases toward them. The model predicts that this channel should generate switching across agro-dealers, especially where ratings reveal meaningful cross-dealer differences. Unlike belief updating about agro-dealer seed in general, dealer sorting requires movement across sellers rather than only increased engagement with agro-dealer seed markets.

We test this prediction by examining whether treated farmers are more likely to switch agro-dealers between adjacent seasons. The analysis is restricted to farmers who purchase maize seed from an agro-input shop in both seasons, since switching is only defined for repeated buyers. Table 10 shows that switching is common: 46% of farmers who purchase maize seed in both baseline and midline buy from a different agro-dealer across the two seasons. The clearinghouse increases switching by 6.2 percentage points at midline, significant at the 10% level. By endline, the estimate is smaller and no longer statistically significant. Exploratory heterogeneity analyses by rating dispersion

Table 9: Treatment effects on farmers' harvest on specific maize plot

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Production in kg [†]	463.203 (399.595)		-20.372 (14.529)	2884	43.937* (17.922)	2898
Area in acres	1.094 (0.655)		-0.003 (0.029)	3004	0.006 (0.038)	3066
Yield in kg/acre [†]	443.013 (304.991)		-23.006 (16.964)	2878	55.606** (17.283)	2891
Production minus cost of seed in UGX ¹	307252.230 (294528.908)		-5960.653 (10229.650)	2443	65621.966** (22927.424)	2525
Amount kept as seed in kg [§]	14.506 (18.530)		-0.188* (0.092)	2931	0.036 (0.104)	2861
Amount sold in kg ^{§†}	195.295 (297.545)		-0.201 (0.124)	3063	0.173 (0.173)	3137
Sales price in UGX/kg	506.954 (139.389)		33.027* (14.244)	610	12.614 (41.238)	639
Revenue in UGX ^{§†}	97.783 (156.538)		-0.393 (0.257)	3058	0.355 (0.363)	3109
Overall Anderson index	-0.020 (0.784)		-0.061 (0.039)	2932	0.097* (0.041)	2900
Max. number of obs.				3407		3441

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

¹Production in UGX, measured as the number of maize bags harvested, multiplied by the market value per bag in UGX, minus the cost of maize seed used in UGX, calculated as the quantity of seed used in kg, multiplied by the cost per kg of seed in UGX.

Table 10: Treatment effects on farmers’ switching behavior

	<i>midline</i>	<i>midline</i>		<i>endline</i>	
	mean	treated	obs.	treated	obs.
Farmer switched to different shop	0.457 (0.498)	0.062 ⁺ (0.033)	1270	0.031 (0.033)	1410
Max. number of obs. ¹			1270		1410

Note: Column (1) reports midline means and standard deviations below (because this variable was not collected at baseline); columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels.

¹Comparisons are restricted to farmers who purchased maize seed from an agro-input shop for any plot at mid- or endline and during the season before that (i.e., base- or midline).

provide similarly limited support for dealer sorting: switching appears, if at all, only in catchment areas with unusually dispersed dealer ratings.

These results provide only weak evidence for dealer sorting as a primary channel. The intervention may have induced some short-run reallocation across sellers, but the effect is not persistent. Moreover, switching alone does not establish that farmers sorted toward higher-performing dealers, especially because the adoption and business-activity results point to broader engagement with agro-dealer seed markets. This interpretation is consistent with [Hasanain, Khan, and Rezaee \(2023\)](#), who find no evidence that livestock farmers switch to higher-performance veterinarians.

6.4.2 Supply-side accountability

We next examine the supply-side accountability channel. In the model, public ratings increase the reputational return to selling well-performing seed. This channel has two empirical implications. First, dealers may respond when ratings make quality more visible, including through seed handling and storage practices, customer service, or quality signaling. Second, if these responses improve the seed itself, they should translate into measurable improvements in dealer-level seed performance. By contrast, belief updating changes farmers’ perceptions and adoption decisions without changing seed performance at a given dealer. Evidence from other contexts shows that public quality disclosure can induce sellers to adjust behavior ([Jin and Leslie, 2003](#); [Klein, Lambertz, and Stahl, 2016](#); [Annan, 2025](#)).

Table 11 reports effects on agro-dealer practices. We construct two indices from enumerator observations of seed handling and storage: labor-intensive practices, such as not storing seed in open containers, and capital-intensive practices, such as having a leak-proof roof. Neither of the two indices responds to the intervention. Thus, we find little evidence that the clearinghouse induced dealers to change the storage and handling practices most directly related to seed preservation.

The intervention does, however, generate suggestive evidence of customer-facing

dealer responses. We measure service provision from both dealer and farmer reports. The dealer-reported service index, including recommendations about complementary inputs, is higher among treated dealers at endline; a parallel farmer-reported service index is higher for treated dealers at midline. Although the timing differs, both measures point to improved customer-facing service provision. We also find a positive effect on activities that signal credibility or professionalism to customers, such as membership in a professional association: a fifth index capturing signaling practices is higher among treated dealers at endline. These patterns suggest that dealers responded to increased transparency, especially along service and signaling margins.

We then examine whether these dealer responses translated into measurable improvements in seed performance. Table 12 reports treatment effects on seed-performance indicators. Enumerators purchased an original, sealed maize seed bag from agro-dealers and evaluated moisture content, packaging integrity, and packaging information, including the visibility of the packaging date and lot number and the number of days since packaging. We find no significant treatment effects on these indicators.

Because not all sampled agro-dealers had maize seed in stock at the time of the survey, the seed-performance sample is smaller than the main dealer sample and may not fully capture the seed available to farmers at planting. In particular, if higher-performing dealers sold out earlier, or if the relevant quality margin occurred for seed purchased earlier in the season, bag-level measures collected at survey time could understate supply-side improvements. To assess whether the null result is driven by limited precision, Appendix A.7 reports a power analysis for seed moisture content, the pre-registered objective measure. The analysis shows that the observed endline effect is small relative to baseline variation and would remain difficult to detect even with substantially larger samples. As a complementary check, we examine additional seed-performance proxies available for larger samples and again find no treatment effects. We therefore interpret the seed-performance results as evidence against detectable improvements in the measures available in our data, rather than as ruling out all supply-side quality responses.

Taken together, the supply-side results show some evidence that dealers responded to increased transparency, primarily through customer-facing service and signaling margins. However, we do not find detectable improvements in seed handling, storage, or measured seed performance. The evidence therefore supports a limited supply-side response, but not the stronger version of supply-side accountability in which reputational pressure improves the underlying performance of seed sold by agro-dealers.

7 Conclusion

This study examines how aggregated peer ratings of seed performance affect farmer beliefs, agro-dealer choice, and agro-dealer behavior. The intervention could operate through three channels: correcting pessimistic or imprecise beliefs among farmers (be-

Table 11: Treatment effects on agro-dealers' practices

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Index of labor-intensive seed handling practices ^{1†}	0.010 (0.484)		0.099 (0.065)	285	0.074 (0.068)	274
Index of capital-intensive seed handling practices ^{2†}	0.000 (0.559)		0.000 (0.072)	270	0.057 (0.083)	265
Index of services (self-reported by agro-dealers) ^{3†}	0.000 (0.382)		0.066 (0.060)	243	0.110* (0.052)	247
Index of services (reported by farmers) ^{4†}	-0.027 (0.583)		0.301** (0.069)	259	0.086 (0.084)	271
Index of quality signaling practices ^{5†}	0.002 (0.686)		0.153 (0.103)	267	0.265+ (0.147)	255
Overall Anderson index	0.034 (0.537)		0.375** (0.113)	162	0.178+ (0.106)	177
Max. number of obs. for dealer survey outcomes				306		297

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

¹The index of labor-intensive seed handling practices contains 6 variables: whether seed is stored in dedicated area, in correct lighting, on correct surface, not in open containers, whether shop has no pest problem, cleanliness and professionalism rating by enumerator.

²The index of capital-intensive seed handling practices contains 5 variables: whether roof is leak-proof, roof is insulated, walls are insulated, shop is ventilated, expired seed is handled correctly. All variables were observed or verified by enumerators, except the last, which is self-reported.

³The index of services (self-reported) contains 8 variables: whether shop offers explanations, complementary input recommendations, extension/training, discounts for larger quantities, credit, did not receive seed related customer complaint, accepts mobile money, stocks small (not repackaged) seed package.

⁴The index of services (reported by farmers) contains 7 variables: whether shop offers refund/insurance, credit, training/advice, delivery, after-sales service, accepts different payment methods, sells small quantities. Farmers' responses are aggregated at dealer level, then the index is computed.

⁵The index of quality signaling practices includes 5 variables: whether shop displays official certificate, is registered with UNADA, is a member of another professional association, holds a trading license issued by local government, number of shop inspections.

Table 12: Treatment effects on agro-dealers' seed performance indicators

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Moisture in % [†]	13.564 (1.482)		-0.122 (0.144)	175	-0.220 (0.197)	261
Bag shows packaging date [†]	0.689 (0.464)		0.050 (0.072)	179	0.035 (0.064)	265
Shelf-life in days ^{1†}	60.951 (40.960)		-8.272 (20.869)	164	6.352 (8.289)	240
Seed is in original undamaged bag [†]	0.940 (0.238)		0.002 (0.046)	179	0.051 (0.055)	265
Bag shows lot number [†]	0.508 (0.501)		-0.001 (0.107)	179	0.027 (0.064)	265
Overall Anderson index	0.065 (0.364)		0.108 (0.103)	160	0.108 (0.090)	236
Max. number of obs. ²				179		265

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; [†] indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

¹Days since the packaging date or, if the bag does not show the packaging date, days since the expiry date minus 6 months.

²The comparisons were only made for agro-dealers where the enumerator was able to buy a bag of maize seed at mid- or endline. Also, we do not control for the baseline values of the outcome variables in the entire table because only 144 of the 179 dealers who had seed at midline also had seed at baseline and only 183 of the 265 dealers who had seed at endline also had seed at baseline, so that controlling for baseline values would reduce the sample sizes drastically.

belief updating), helping farmers identify and switch to better-performing dealers (dealer sorting), or inducing agro-dealers to invest in seed preservation (supply-side accountability). Using a randomized experiment, we evaluate the effects of this intervention on a range of farmer and agro-dealer outcomes and distinguish between these channels.

We find that the intervention leads farmers to update their beliefs and improves their perceptions of seed performance. In an exploratory analysis, we find that effects on perceptions are larger among farmers without prior purchasing experience, consistent with the model’s prediction that farmers facing greater uncertainty about seed performance weight the peer signal more heavily.

As farmers update their beliefs, the intervention increases their adoption of high-yielding maize varieties purchased from agro-dealers and leads to higher productivity and improved harvest outcomes. The combination of positive treatment effects on perceptions, adoption, and productivity suggests that prior beliefs were overly pessimistic relative to seed performance and that the seeds accessed through agro-dealers performed favorably compared with outside options. Consistent with the model, these findings indicate that aggregated peer ratings can move farmers’ beliefs closer to the performance of seed sold through agro-dealers, shifting adoption decisions toward the welfare-optimal level.

This finding is consistent with existing evidence. Studies from fertilizer markets show that farmers often underestimate input quality relative to objective measures (Michelson et al., 2021; Hoel et al., 2024), and recent work on maize seed in Uganda suggests that farmers’ perceptions may be more pessimistic than warranted by measured performance (Barriga and Fiala, 2020). Insights from psychology and economics further suggest that negative information tends to be more salient, spreads faster, and receives disproportionate weight in social learning under noisy production conditions, which can skew belief formation toward pessimism (Rozin and Royzman, 2001; Vaish, Grossmann, and Woodward, 2008; Conley and Udry, 2010; Hornik et al., 2015).

On the supply side, the evidence is more limited. We find no improvement in seed handling, storage practices, or objective seed quality. There is suggestive evidence of some changes in service provision and signaling professionalism, but these effects are less robust than the demand-side findings and are not central to our interpretation. While we find some evidence that the intervention induces switching between agro-dealers, this switching is not directed toward higher-rated dealers, providing little support for dealer sorting as a primary channel. The pattern of results — positive effects on perceptions and adoption, undirected switching and no improvement in seed quality — points to belief updating as the primary impact pathway.

This study highlights the potential for aggregated peer information to improve market outcomes in environments where buyers face uncertainty about input performance and learning from own experience and social networks is limited. By pooling experiences from multiple users, such signals can help farmers form more accurate beliefs about input performance.

More broadly, our findings suggest that the primary channel through which crowd-sourced information platforms affect market outcomes depends on the nature of the

binding constraint. In markets where consumers already participate but face difficulty choosing among sellers, the main channel is sorting — as documented in the online reviews literature. In markets where the binding constraint is participation itself — as in our setting, where many farmers do not engage with agro-dealers due to uncertainty about seed performance — the main channel is belief updating at the extensive margin. This distinction has implications for how clearinghouse systems should be designed: when the goal is to increase market participation rather than improve matching, broad dissemination to non-participants matters more than fine-grained seller rankings.

These findings have implications for policy and practice. Aggregating and disseminating users’ experiences can improve market outcomes when quality assurance systems are weak or imperfect, offering a decentralized complement that does not rely on strong institutional capacity. Peer ratings help farmers form more accurate beliefs, reducing the barrier to adoption. There is also suggestive evidence that the accountability pressure from ratings may incentivize service improvements among agro-dealers, though this channel requires further investigation. For private actors, including agro-dealers and seed companies, experience-based feedback mechanisms can build credibility and strengthen market engagement.

At the time of implementation, limited smartphone access required an offline design. As digital access expands among smallholder farmers, similar systems could be implemented through online platforms, offering lower costs, broader reach, and greater scalability, although this remains to be tested.

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A Appendix

A.1 Context on seeds and their quality

In this study, the term “high-yielding maize varieties” refers to both open-pollinated varieties (OPVs) and hybrids that have been genetically improved through public or private sector breeding programs. While the difference between these two types of maize is based on their reproductive biology, there are also economic implications ([Morris, Rusike, and Smale, 1998](#); [Spielman and Kennedy, 2016](#)). We explain the terminology and our rationale for using this terminology below, recognizing that our explanation is a simplification of much more complex biological phenomena.

Maize hybrids tend to confer a significant advantage in terms of vigor – robust growth, high yields, and uniform yield performance, even under suboptimal conditions. This is due to the expression of heterosis that results from crossing distinct inbred parent lines to produce genetically superior progeny (F1) that is planted by the farmer. However, this vigor declines dramatically if farmers save and plant grain that was set aside from harvest. Instead, the farmer must purchase fresh (F1) seed each season to continue to realize the benefits of heterosis. The combination of vigor and purchase frequency can incentivize public investment in hybrid maize improvement, seed production, and marketing.

OPVs, on the other hand, are typically less vigorous than hybrids. Although they are also less susceptible to genetic degeneration and can be saved and planted across several generations without loss of vigor, their yield performance is typically lower than hybrids. Moreover, OPV seed saving practices – and thus less frequent purchases of fresh seed – tend to reduce these private investment incentives.

Relatedly, we use the term “farmer-saved seed” in this study to refer to seed that is of an unknown progeny or quality. They may be seed saved from the harvest of F1 hybrids or OPVs, and they may be n generations old. They may have been stored under suboptimal conditions between seasons, exposed to moisture, pests, or diseases, and, as a result of both genetic and physical factors, perform poorly relative to fresh hybrid or OPV seed purchased from an agro-input shop.

As highlighted in a recent review by [Michelson et al. \(2023\)](#), seed quality is complex and multi-dimensional due to the living nature of seeds, encompassing three essential dimensions. First, analytical purity measures the proportion of a seed sample that belongs to the intended species. Second, varietal purity ensures the seed is of the expected variety, possessing the claimed genetic characteristics. Third, viability and performance assess the seed’s capacity to germinate and grow with sufficient vigor.

This study focuses on the viability and performance dimension of maize seed quality. From a farmer’s perspective, this dimension is particularly salient, as it represents a crucial outcome that farmers observe during germination, early crop growth, and harvest, although it can be difficult to attribute realized outcomes to seed performance alone. Moreover, while some storage and handling practices, such as seed mixing or mislabeling, may affect varietal purity, the practices examined in this study primarily

influence seed viability and performance.

Objectively assessing maize seed performance is complex, but measuring the moisture content of a seed sample is the most common and practical test. Moisture is a powerful predictor, as it directly affects germination outcomes. [Afzal et al. \(2017\)](#), e.g., report germination rates of approximately 85% for seeds with a moisture content of 12.7%, compared to just 50% for the same seeds at 15.6%. Prolonged exposure to high humidity rapidly degrades seed viability by increasing susceptibility to damage from pathogens, insects, and physical handling ([Michelson et al., 2023](#)). The moisture content of a seed sample reflects recent storage conditions – whether the seeds were kept in airtight packages or exposed to open air. Consulted experts emphasized a widely used rule of thumb: each one-percentage-point increase in moisture content halves the seed’s shelf life.

While moisture measurement is standard practice among seed companies and crop inspectors and provides a valuable indication of seed performance, it is inherently one-dimensional. Additional quality assessments include germination tests conducted in controlled chambers; vigor tests under the growing conditions that may occur in the field; grow-out tests, which evaluate genetic purity through morphological analysis; seed health testing which measures the presence of pathogens and diseases, and genotyping (DNA fingerprinting), which verifies genetic purity using single nucleotide polymorphism analysis.

A.2 Rating computation

What if a treated agro-dealer does not receive a single rating? If a treated agro-dealer is not rated by any farmer – e.g., because no sampled farmer had personally purchased seed there or knew someone who had – we informed farmers that we had no information about the shop while still mentioning its existence. In the first round, 16 out of 193 treated agro-dealers were not rated by any farmer.

Should more ratings lead to better ratings? Some agro-dealers were not rated by any farmer in the first round, while others received up to 22 ratings. If agro-dealer A is rated by ten farmers with an average of 3.5 and agro-dealer B is rated by one farmer with a rating of 3.6, agro-dealer B is treated as having the higher rating.

Should ratings reflect catchment area performance? Ratings were highly concentrated at the upper end of the scale, with fewer than 9% of agro-dealers receiving a rating below three out of five. Ratings are therefore transformed into quintiles based on their distribution. To avoid conveying an overly negative signal, the quintile labels were chosen to carry positive connotations.

Are female agro-dealers rated worse than male agro-dealers? Significant differences were observed between the ratings of female agro-dealers (41% of the sample) and male agro-dealers (59%), even after controlling for potential confounders such as education and multiple performance indicators. These differences are unlikely to reflect true disparities in performance and instead appear to stem from biased perceptions, likely driven by discrimination (De, Mieke, and Van Campenhout, 2024). To mitigate potential harm from the intervention, the ratings of female agro-dealers were adjusted accordingly. Specifically, each seed performance attribute was regressed on a gender dummy, and the resulting coefficients were added to the initial ratings of female agro-dealers.

A.3 Agro-dealer Rating Report



Pendaya's Agro Shop
scores

2.4 out of 5



This was based on 5 reviews.

This score means that farmers in the area think that the
quality of the maize seed this shop sells is:

Good!

SeedAdvisor certificate 2022



Hi there!

As a business, you know that your customers' opinion is very important! Your sales and profits depend on what customers think about the quality of the products you sell.

We visited farmers in your area and asked them about their experience with maize seed that they bought in shops such as yours. We have used this information to compute a score for you. In this way, you get an idea of what customers think about the maize seed you sell, and how it compares to other agro-input shops in your area. This gives you the opportunity to improve your business even more!

The overall score indicated on the certificate combines different components. You can find separate ratings in the table below. For instance, we asked what customers think of maize seed quality in general. But we also asked customers about their experiences with respect to yield, drought and disease resistance, time to mature, and germination, keeping in mind what was advertised when they bought the seed.

Check your scores carefully and see where you can do better! You can also compare your own results to what customers think of the quality of maize seed sold by other agro-input dealers in your area.

	you	other input dealers
What do customers say about overall seed quality?	★★★★☆☆	★★★★☆☆
Do customers think yield of maize seed is as advertised?	★★★★☆☆	★★★★☆☆
Do customers think maize seed is as drought resistant as advertised?	★★★★☆☆	★★★★☆☆
Do customers think maize seed is as disease resistant as advertised?	★★★★☆☆	★★★★☆☆
Do customers think maize seed matures as advertised?	★★★★☆☆	★★★★☆☆
What do customers think about maize seed germination?	★★★★☆☆	★★★★★★

Tip: Display this certificate in your shop to inform customers about your score. You will be scored again in July 2022, so keep standards up!

SeedAdvisor is a service of Digital Natives LLC in collaboration with the International Food Policy Research Institute. Ratings are based on crowdsourced data that is aggregated to protect the privacy of the raters. The service and data is provided as is, and makes no warranties, either expressed or implied, concerning the accuracy, completeness, reliability, or suitability of the information.

A.4 Differential attrition tests

To examine differential attrition, we regress the pre-registered balance variables (see Table 4) on a binary variable indicating whether an agro-dealer left the sample, a binary variable indicating whether an agro-dealer belonged to the control group, and their interaction. Column (2) reveals that agro-dealers who left the sample at midline operated closer to the nearest tarmac road – the only significant difference between those who left and those who stayed when attrition was most pronounced (see Table 5). By endline, however, those who left were younger, more likely to be female, had less established businesses, were more likely to operate specialized shops, and still tended to be closer to a tarmac road, see column (5). This suggests that agro-dealers who left differed from those who remained, regardless of exposure to the treatment, possibly reflecting greater vulnerability during the COVID-19 period, leading to business closures. Columns (3) and (6) show minimal differences between control and treated agro-dealers, consistent with Subsection ???. Examining the interaction between the indicators reveals few differences between agro-dealers who left the sample *and* belonged to the control group and those who did not, suggesting that differential attrition is unlikely to bias our estimates.

Table A1: Differential attrition tests - Agro-dealer

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean	attrited	belongs to control	attrited and control	belongs to control	attrited and control
Respondent's age in years	32.43 (11.49)	-2.50 (2.25)	2.68 ⁺ (1.40)	-1.29 (3.04)	1.59 (1.40)	5.45 ⁺ (3.04)
Resp. is male	0.59 (0.14)	-0.17 (0.06)	0.05 (0.17)	-0.12 (0.49)	-0.34 ^{**} (0.10)	0.28 ⁺ (0.14)
Resp. finished primary education	0.92 (0.27)	0.01 (0.08)	0.01 (0.04)	-0.02 (0.10)	0.04 (0.05)	-0.04 (0.08)
Resp. owns shop	0.55 (0.50)	-0.15 (0.17)	-0.01 (0.06)	0.01 (0.20)	-0.05 (0.16)	0.14 (0.20)
Resp. received training on maize seed handling	0.53 (0.50)	0.14 (0.11)	-0.11 (0.07)	-0.18 (0.16)	0.10 (0.10)	-0.19 (0.15)
Resp. knows how to store seed after repackaging	0.27 (0.44)	0.13 (0.17)	-0.08 (0.06)	-0.10 (0.19)	0.03 (0.12)	0.03 (0.15)
Dealer's distance to nearest tarmac road in km	6.56 (10.39)	-5.37 ^{**} (1.37)	1.68 (2.33)	2.63 (2.63)	-3.38 ⁺ (1.74)	0.92 (2.93)
Dealer only sells farm inputs	0.74 (0.44)	0.11 (0.10)	0.00 (0.07)	-0.18 (0.14)	0.23 ^{**} (0.07)	-0.27 [*] (0.12)
Years since dealer establishment	5.34 (6.30)	-1.93 (1.26)	0.07 (0.81)	-0.35 (1.78)	-2.99 [*] (0.96)	0.60 (1.51)
Number of customers per day	41.49 (46.49)	-7.99 (8.66)	-5.02 (7.50)	-5.48 (10.59)	-8.05 (7.01)	-4.79 (9.53)
Quantity of maize seed sold in kg	695.50 (1497.18)	234.18 (833.31)	-144.71 (376.10)	-847.11 (875.12)	125.10 (459.81)	-551.67 (483.08)
Maize seed lost/wasted last season in kg	3.50 (18.65)	2.73 (8.06)	-2.22 (2.35)	-3.06 (8.16)	3.91 (9.08)	-5.89 (9.21)
Dealer has problem with pests	0.65 (0.48)	0.16 (0.11)	0.08 (0.06)	-0.36 [*] (0.17)	-0.02 (0.09)	-0.08 (0.14)
Dealer stores maize seed in open containers	0.16 (0.36)	-0.05 (0.10)	-0.09 ⁺ (0.05)	0.09 (0.12)	-0.05 (0.10)	0.09 (0.12)
Dealer received seed related customer complaint	0.64 (0.48)	-0.03 (0.12)	-0.04 (0.05)	-0.14 (0.17)	-0.11 (0.12)	0.05 (0.16)
Moisture in bag of maize seed in %	13.56 (1.44)	-0.42 (0.52)	0.10 (0.27)	0.27 (0.66)	-0.45 (0.31)	0.48 (0.56)

Note: Column (1) reports sample means at baseline and standard deviations below; columns (2) and (5) report differences between agro-dealers who attrited and those who did not, and standard errors below; columns (3) and (6) report differences between control and treatment agro-dealers, and standard errors below; columns (4) and (7) report differences between agro-dealers who attrited and belong to the control group and those who do not, and standard errors below; all standard errors are clustered at the level of randomization; **, *, and + denote significance at the 1%, 5%, and 10% levels.

A.5 Agro-dealer rating patterns

Comparison of rating dimensions We asked farmers to evaluate maize seed sold by an agro-dealer across six dimensions: General quality, Yield as advertised, Drought tolerance as advertised, Pest/disease tolerance as advertised, Speed of maturing as advertised, and Germination. Figure A1 presents the distribution of farmers’ ratings for these attributes along with their mean values.

The figure reveals variation across attributes, with some showing more concentrated distributions and others exhibiting greater dispersion. The attributes most strongly correlated with the *General quality* rating are *Yield as advertised* (0.54) and *Germination* (0.38). These two attributes also have the smallest mean absolute differences (0.47 for *Yield as advertised* and 0.55 for *Germination*), suggesting that these are the attributes farmers primarily consider when assessing overall performance.

Comparison of ratings by purchase channel Farmers could rate an agro-dealer if they met one of two conditions: either they had purchased seed from the agro-dealer themselves or they knew someone who had. Ideally, only farmers with firsthand experience would provide ratings, but restricting responses in this way risked generating too few observations for a reliable signal. While this decision was made at the experiment’s design stage and cannot be revised ex post, we can compare endline ratings between farmers who purchased seed from the agro-dealer and those who did not but knew someone who had.

Fortunately, the vast majority (93%) of ratings came from farmers who had purchased seed, while only 7% were provided by those without direct purchase experience. This alleviates concerns that ratings are primarily driven by reputation or hearsay effects. While the mean ratings differ slightly (3.6 for farmers who bought seed vs. 3.7 for those who did not), the density plots are similar (see Figure A2).

Correlating ratings and performance indicators There is concern that small-holder farmers may base their ratings of agro-input shops on factors such as customer service rather than seed performance or rely on the agro-dealer’s reputation or personal relationship with them. To address this, we test whether the ratings – specifically, the index of endline ratings computed following [Anderson \(2008\)](#) – correlate with objective indicators of seller and seed performance, also measured at endline.

We find that specialized agro-input shops, which exclusively sell farm inputs, receive higher ratings, as do dealers with better seed handling practices, objectively observed by our enumerators (Table A2). Also, farmers not only rated seeds purchased from agro-input shops but also those used on randomly selected maize fields, using the same questions. These seed ratings are positively correlated with crop yields (Table A3).

While alternative explanations for these significant correlations exist, the overall pattern supports our claim that farmers are assessing seed performance. The ratings therefore appear to provide an informative signal, although farmers’ perceptions may not be perfectly accurate given the many factors influencing agricultural production.

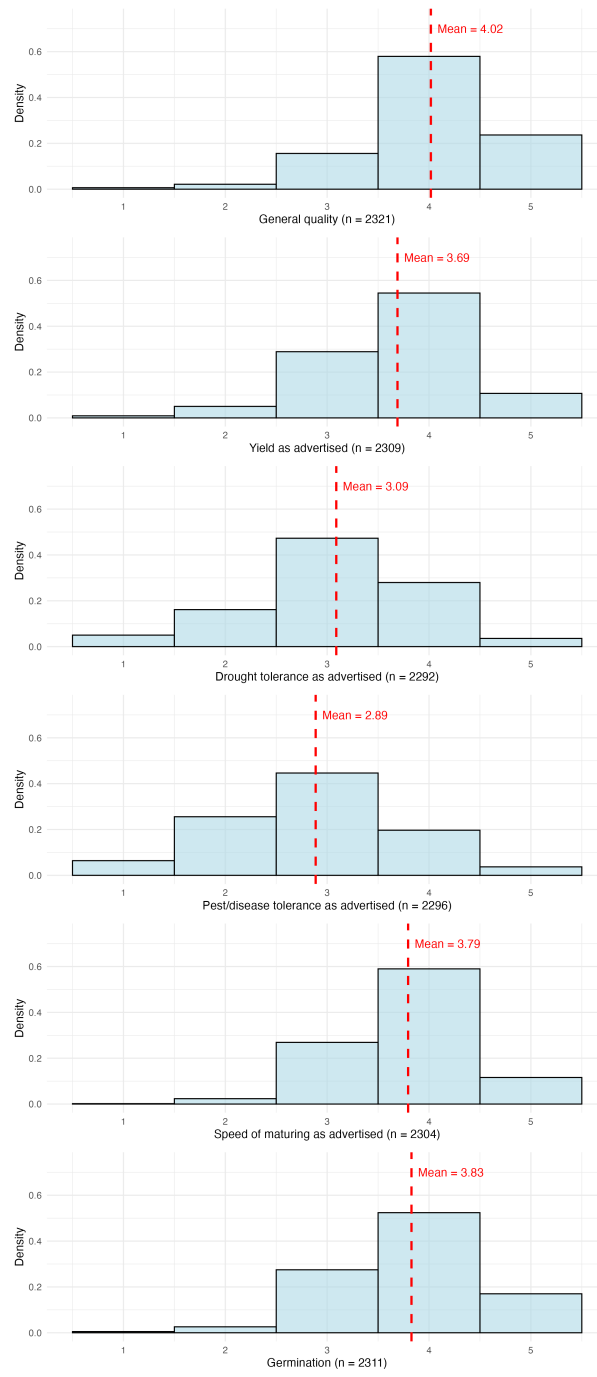


Figure A1: Comparison of rating dimensions at endline

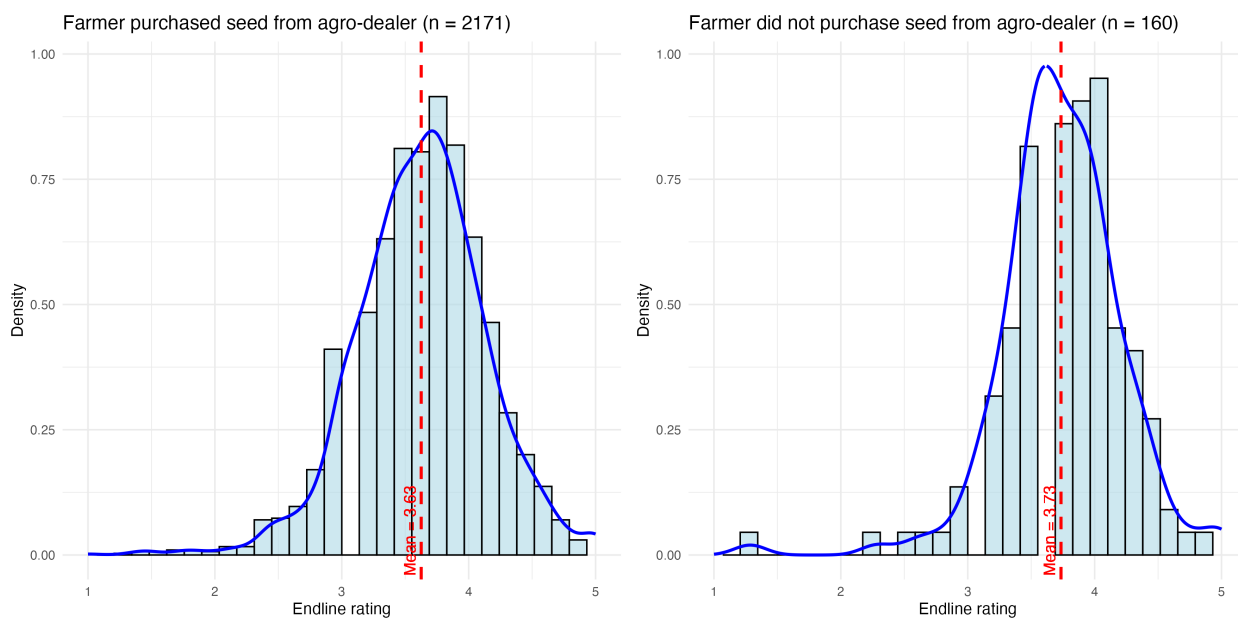


Figure A2: Comparison of ratings by purchase channel

Table A2: Correlating ratings and performance indicators - Agro-dealer level

	Ratings
Agro-dealer only sells farm inputs	0.205** (0.078)
Index of labor-intensive seed handling practices ¹	0.164+ (0.091)
Index of capital-intensive seed handling practices ²	0.156+ (0.081)
Index of all seed handling practices ³	0.205+ (0.116)
Agro-dealer received seed related complaint from customer	-0.108 (0.093)
Shop received warning after inspection	-0.034 (0.086)

Note: ¹The index of labor-intensive seed handling practices contains 6 variables: whether seed is stored in dedicated area, in correct lighting, on correct surface, not in open containers, whether shop has no pest problem, cleanness and professionalism rating by enumerator.

²The index of capital-intensive seed handling practices contains 6 variables: whether roof is leak-proof, roof is insulated, walls are insulated, shop is ventilated, shop displays official certificate, expired seed is handled correctly.

³The index of all seed handling practices contains 12 variables: the ones included in the index of capital-intensive practices and the ones included in the index of labor-intensive practices.

Table A3: Correlating ratings and yield on randomly selected field - Farmer level

	Yield in kg/acre _{endline}
Index of farmer's seed ratings _{midline} ¹	144.971** (8.564)
Index of farmer's seed ratings _{endline} ¹	35.225** (9.484)

Note: ¹The index of farmer's seed ratings contains 6 ratings: general quality, yield, drought tolerance, pest/disease tolerance, speed of maturing, and germination.

A.6 Treatment effects on agro-dealers' business activity

We examine treatment effects on agro-dealer business activity for two maize varieties: the most recently released hybrid, Longe 10H (Table A4) and the most recently released OPV, Longe 5 (Table A5). In addition to the general business outcomes described in Subsection 6.2, we include stock indicators. Agro-dealers reported how much seed of each variety was carried over from the previous season, how much was procured from suppliers, how much was lost or wasted, and how often stock-outs occurred.

Higher demand induced by the intervention should increase turnover, reducing carryover and increasing procurement of fresh seed. Higher turnover may also reduce losses and stock-outs.

At midline, we find no significant effects. By endline, however, all coefficients move in the expected direction, and the combined [Anderson \(2008\)](#) index shows a positive and statistically significant effect. This pattern is consistent with increased farmer engagement translating into higher agro-dealer business activity.

Table A4: Treatment effects on agro-dealers' business activity (Longe 10H)

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Sales volume in kg ^{§†}	288.384 (727.049)		0.236 (0.204)	256	0.352 (0.239)	242
Sales price in UGX/kg ^{§†}	6220.899 (1153.371)		-0.013 (0.026)	194	0.010 (0.029)	187
Revenue in mln UGX ^{§†}	1.625 (3.839)		0.130 (0.123)	255	0.173 (0.136)	241
Amount carried over in kg ^{1§†}	2.679 (12.137)		0.090 (0.215)	262	-0.034 (0.134)	250
Amount shop bought from provider in kg ^{§†}	294.672 (741.810)		0.206 (0.213)	257	-0.022 (0.253)	243
Amount lost/wasted in kg ^{§†}	0.036 (0.405)		0.019 (0.097)	257	-0.038 (0.041)	243
Number of times per month shop ran out ^{§†}	1.039 (1.575)		-0.045 (0.133)	192	-0.205 (0.136)	185
Overall Anderson index	0.080 (0.437)		0.029 (0.070)	244	0.217** (0.057)	233
Max. number of obs. ²				268		254

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

¹Amount carried over is low at baseline, as many agro-dealers report carrying no seed over from the previous season.

²The comparisons were only made for agro-dealers that had Longe 10H in stock at mid- or endline.

Table A5: Treatment effects on agro-dealers' business activity (Longe 5)

	<i>baseline</i>		<i>midline</i>		<i>endline</i>	
	mean		treated	obs.	treated	obs.
Sales volume in kg ^{§†}	389.492 (716.556)		0.304 (0.216)	261	0.316 (0.230)	259
Sales price in UGX/kg ^{§†}	3114.463 (409.328)		-0.015 (0.016)	249	0.013 (0.022)	241
Revenue in mln UGX ^{§†}	1.193 (2.175)		0.111 (0.096)	261	0.114 (0.105)	258
Amount carried over in kg ^{1§†}	4.312 (19.088)		-0.092 (0.306)	270	-0.004 (0.155)	263
Amount shop bought from provider in kg ^{§†}	431.451 (803.696)		0.253 (0.215)	262	0.289 (0.235)	260
Amount lost/wasted in kg ^{§†}	1.756 (10.173)		0.031 (0.128)	266	-0.055 (0.058)	261
Number of times per month shop ran out ^{§†}	0.839 (1.509)		0.086 (0.101)	248	-0.054 (0.126)	237
Overall Anderson index	0.039 (0.401)		0.012 (0.062)	256	0.152* (0.058)	252
Max. number of obs. ²				275		269

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

¹Amount carried over is low at baseline, as many agro-dealers report carrying no seed over from the previous season.

²The comparisons were only made for agro-dealers that had Longe 5 in stock at mid- or endline.

A.7 Treatment effects on agro-dealers' seed performance

Moisture content – a pre-registered, objectively measured indicator of seed performance – shows a statistically insignificant negative treatment effect. Because not all agro-dealers had maize seed in stock, the realized sample size is smaller than planned: 232 observations at baseline, 179 at midline, and 265 at endline, compared with a target of 348.

To assess whether the null result reflects a true absence of effect or insufficient statistical power, we assume that the estimated treatment effect at endline – a 0.22 percentage point reduction in seed moisture content – represents the true effect. Figure A3 plots a power curve relating sample size to the probability of detecting this effect at the 5% significance level, given the observed baseline variability in moisture content. The curve shows that the null result is unlikely to be driven by the reduced sample size: even with substantially larger samples, an effect of this magnitude would remain undetectable.

We also examine additional indicators of seed performance, including binary variables for whether an agro-dealer received a seed-related customer complaint, a warning following an inspection, or had products confiscated during an inspection. These outcomes are available for larger samples than the indicators derived from purchased seed bags in Table 12. We find no treatment effects on these measures, providing further evidence that the intervention does not affect seed performance.



Figure A3: Power curve for seed moisture at endline

Table A6: Treatment effects on agro-dealers' seed performance (alternative proxies)

	<i>baseline</i>	<i>midline</i>		<i>endline</i>	
	mean	treated	obs.	treated	obs.
Shop received seed related complaint from customer [†]	0.644 (0.480)	-0.089 (0.057)	306	-0.039 (0.048)	297
Shop received warning after inspection [†]	0.317 (0.466)	0.005 (0.073)	291	-0.009 (0.063)	284
Shop's products were confiscated after inspection [†]	0.145 (0.353)	-0.027 (0.046)	293	-0.025 (0.036)	285
Overall Anderson index	-0.007 (0.680)	0.066 (0.100)	303	0.060 (0.087)	292
Max. number of obs.			306		297

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; [†] indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

A.8 Alternative index construction

We recompute all summary indices following [Kling, Liebman, and Katz \(2007\)](#) as a robustness check to the [Anderson \(2008\)](#) indices used in the main analysis. For farmers' perceptions of seed performance, the treatment effect becomes statistically insignificant when the index is constructed following [Kling, Liebman, and Katz \(2007\)](#). Results for farmers' adoption and sourcing decisions remain robust. Results for agro-dealers' business activity (general, Longe 10H, and Longe 5) are largely robust.

For farmers' harvest outcomes, the positive endline effect remains robust. At mid-line, however, the [Kling, Liebman, and Katz \(2007\)](#) index yields a significant negative coefficient. The harvest outcomes included in the index are mechanically related (e.g., yield, production, and revenue). The [Anderson \(2008\)](#) method down-weights highly correlated components, whereas [Kling, Liebman, and Katz \(2007\)](#) assigns equal weights and treats these outcomes as independent, which can amplify the influence of redundant information and generate a negative treatment effect. That is why we rely on the [Anderson \(2008\)](#) method, as pre-registered in our [pre-analysis plan](#), and use [Kling, Liebman, and Katz \(2007\)](#) only as a robustness check.

Results for agro-dealers' practices remain robust, as do all null results regarding agro-dealers' seed performance.

Table A7: Anderson (2008) vs. Kling, Liebman, and Katz (2007) indices

	Anderson		Kling et al.	
	<i>midline</i>	<i>endline</i>	<i>midline</i>	<i>endline</i>
Farmers' perceived seed performance		0.092 ⁺ (0.054)		0.088 (0.056)
Farmers' adoption and sourcing decisions	0.087* (0.042)	0.086* (0.039)	0.096* (0.041)	0.076* (0.038)
Agro-dealers' business activity (general)	0.197* (0.092)	0.238** (0.082)	0.189 (0.125)	0.247 ⁺ (0.132)
Agro-dealers' business activity (Longe 10H)	0.029 (0.070)	0.217** (0.057)	0.060 (0.082)	0.232** (0.076)
Agro-dealers' business activity (Longe 5)	0.012 (0.062)	0.152* (0.058)	0.105 (0.089)	0.160 ⁺ (0.081)
Farmers' harvest	-0.061 (0.039)	0.097* (0.041)	-0.092* (0.046)	0.082 ⁺ (0.046)
Agro-dealers' practices	0.375** (0.113)	0.178 ⁺ (0.106)	0.272* (0.118)	0.238* (0.118)
Agro-dealers' seed performance indicators	0.108 (0.103)	0.108 (0.090)	0.065 (0.111)	0.064 (0.087)
Agro-dealers' seed performance (alternative proxies)	0.066 (0.100)	0.060 (0.087)	0.062 (0.103)	0.059 (0.087)

Note: Columns (1) and (2) report differences between treatment and control groups, with standard errors below (clustered at the level of randomization), if the overall indices are constructed following Anderson (2008); columns (3) and (4) report differences between treatment and control groups, with standard errors below (clustered at the level of randomization), if the overall indices are constructed following Kling, Liebman, and Katz (2007); **, *, and + denote significance at the 1%, 5%, and 10% levels; larger indices indicate more desirable outcomes.